

National University of Computer and Emerging Sciences, Lahore Campus



Course: Electronic Devices and Circuits
Program: BS(EE)
Duration: 180 Minutes
Paper Date: 23rd December, 2016
Section: ALL
Exam: Final

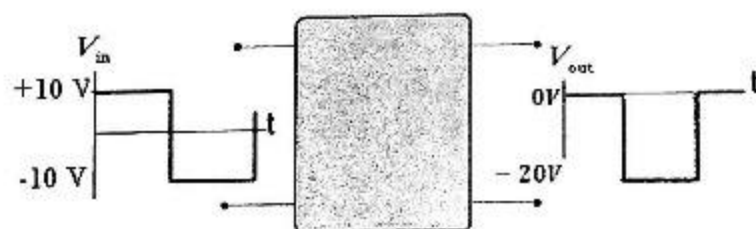
Course Code: EE225
Semester: Fall 2016
Total Marks: 100
Weight: 50%
Page(s): 1
Roll No:

Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together. State your valid assumptions clearly if you have to make any.

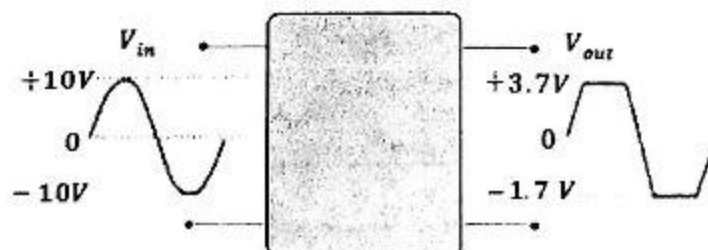
Question # 1 [3+2+2+2+3+2+2+3+3+3=25 marks]

Short Questions:

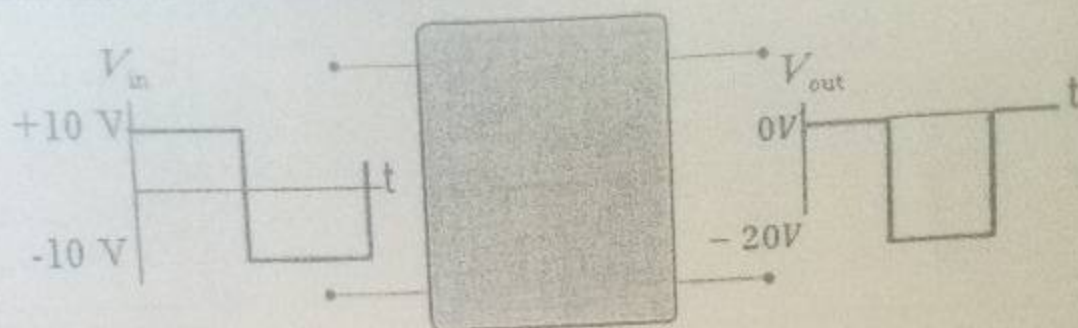
- 1) Draw the block diagram of a dc power supply.
- 2) Define threshold voltage in context of a MOS transistor?
- 3) What is Early voltage?
- 4) Explain channel pinch off in a MOS transistor?
- 5) Using a PNP BJT and voltage sources, draw a circuit in common base configuration such that it works in active mode.
- 6) What is the significance of using emitter resistance in a common emitter amplifier?
- 7) Why coupling capacitors are used in voltage amplifiers?
- 8) Draw the circuit inside the black box whose input and output are given below.



- 9) Draw the circuit inside the black box whose input and output are given below.



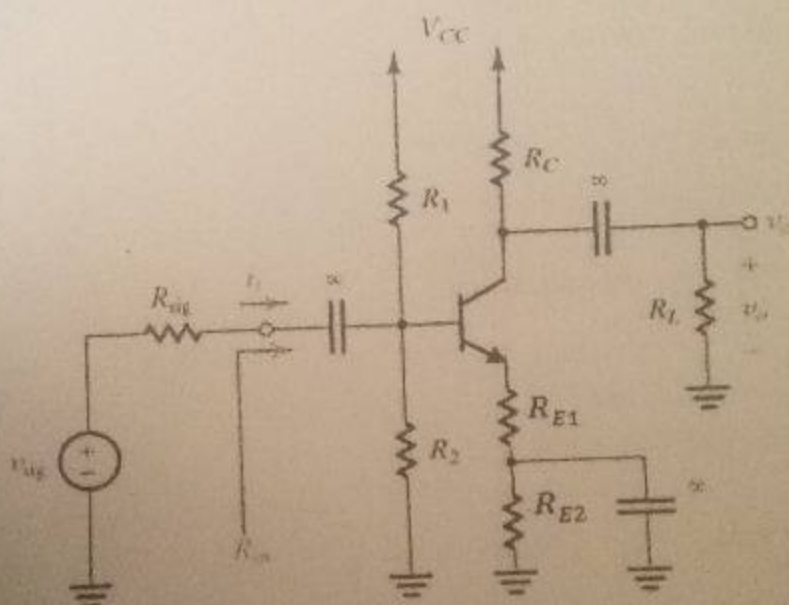
b) Draw the circuit inside the black box.



35 Marks

Question: 3 (CLO # 4,5)

Consider the following amplifier circuit with
 $R_1 = 51k\Omega$, $R_2 = 32.9k\Omega$, $R_C = 4k\Omega$, $R_{E1} = 1k\Omega$, $R_{E2} = 3k\Omega$, $V_{CC} = 12V$
 $R_{sig} = 50\Omega$, $v_{sig} = 1V_{peak}$ and $R_L = 1k\Omega$, $\beta = 100$



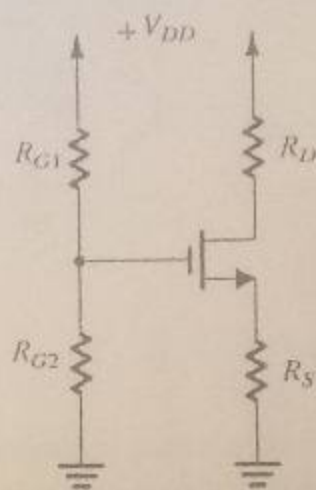
- Find V_B , V_E , V_C , I_E and I_C .
- Draw small signal equivalent model of the circuit.
- Find small signal parameters?
- Find the voltage gain $\frac{v_o}{v_{sig}}$, input resistance and output resistance of the amplifier?
- If we replace both emitter resistors with a single resistor of $4k\Omega$ with capacitor connect to emitter directly, what is the effect on voltage gain, input and output resistances? Verify your results by calculating these parameters?
- Comment which one is a better voltage amplifier (with or without split R_E) and why?

Question: 6 (CLO # 7)

10 Marks

Design the given circuit with following requirements:

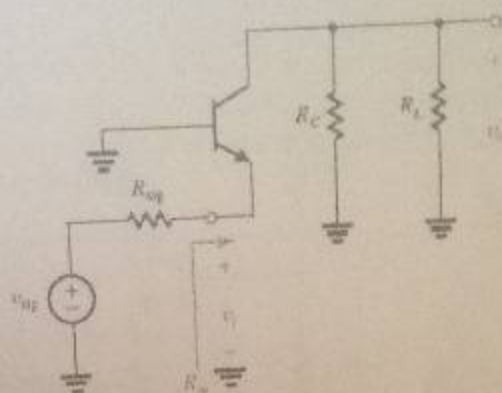
$V_{DD} = +5V$, $I_D = 0.32mA$, $V_S = 1.6V$, $V_D = 3.4V$ with a $1\mu A$ current through the voltage divider R_{G1} , R_{G2} . Let $V_t = 1V$ and $k'_n \left(\frac{W}{L}\right) = 1mA/V^2$. Neglect the channel length modulation effect.



Question: 4 (CLO # 5)

15 Marks

- a) A common base amplifier is required to amplify a signal delivered by a co-axial cable having a source resistance of 50Ω .
- What bias current I_C should be utilized to obtain R_{in} that is matched to the cable resistance?
 - To obtain an overall voltage gain of G_V of 40 V/V, what should the total resistance in the collector be?

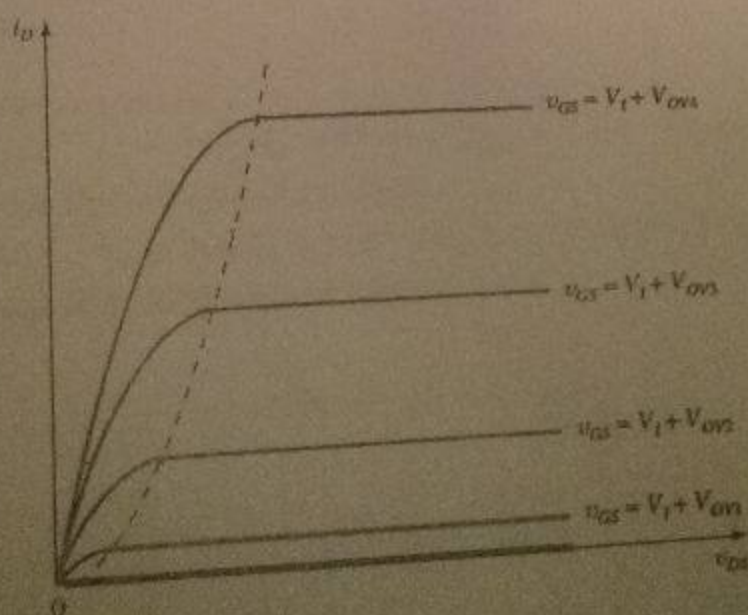


- b) Compare common emitter, common base and common collector amplifiers? Use voltage gain, input resistance and output resistance as analysis parameters.

Question: 5 (CLO # 6)

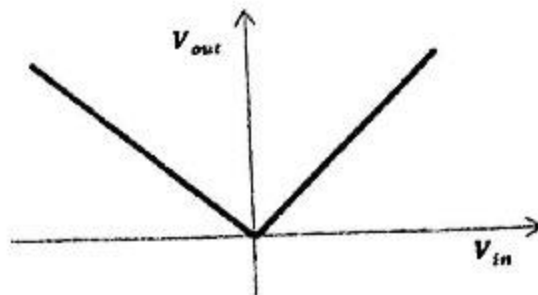
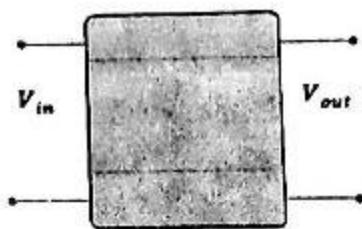
15 Marks

- a) The $i_D - v_{DS}$ characteristics of an n -channel enhancement MOSFET are given in the figure below.
- Identify the three regions of operation of the NMOS on the characteristic curve.
 - Give the Operating conditions for V_{DS} and V_{GS} and expression for current I_D in each region.



- b) Draw and label the physical structure of a PMOS transistor.

10) Draw the circuit inside the black box which has the transfer characteristics as given below.



Question # 2 [15 marks]

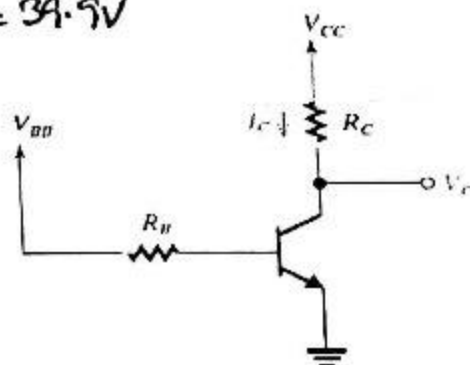
A shunt regulator utilizes a Zener diode whose voltage is 5.1V at a current of 50mA and whose incremental resistance is 7Ω . The diode is fed from a supply of 15V nominal voltage through a 200Ω resistor. What is the output voltage at no load? Find the line regulation and load regulation?

Question # 3 [10 marks]

For the circuit shown in figure below, $V_{CC} = 10V$, $R_C = 10K\Omega$ and $R_B = 200K\Omega$. The BJT has $\beta = 100$. Find the value of V_{BB} that result in transistor operating

- In active mode with $V_C = 2V$ $V_{BB} = 2.3V$
- At the edge of saturation $V_{BB} = 2.64V$
- Deep in saturation with $\beta_{forced} = 20$.

$$V_{BB} = 3.97V$$



Handwritten calculations for Question 3:

$$V_{BB} = 2.3V$$

$$V_{BB} = 2.64V$$

$$V_{BB} = 3.97V$$

$$V_{CC} = 10V$$

$$R_C = 10k\Omega$$

$$R_B = 200k\Omega$$

$$V_C = 2V$$

$$V_E = 0$$

$$V_B = 0.7V$$

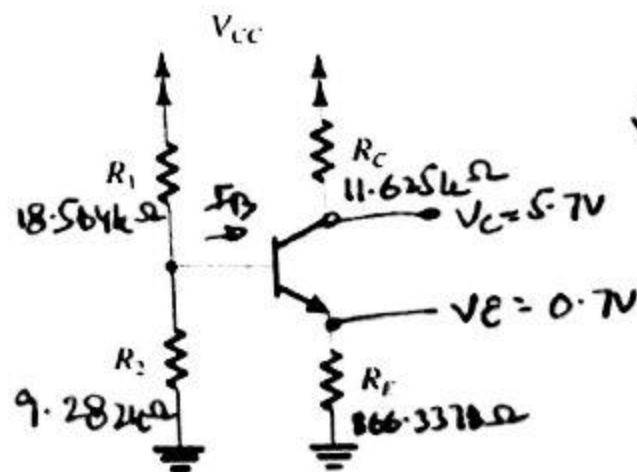
$$10 - 2 = I_C \cdot 10k$$

$$I_C = 0.8mA$$

$$I_C (\beta + 1) = \frac{V_{BB} - V_B}{R_B}$$

Question # 4 [15 marks]

Design the circuit shown in the figure with $\beta = 120$ such that the transistor operates at a Q point of $V_{CE} = 5V$ for $V_{CC} = 15V$, $I_C = 0.8mA$, and the voltage across resistor R_E is 0.7V. Assume the current flowing through potential divider network is equal to the emitter current.



Handwritten calculations for Question 4:

$$V_{BB} = 5V$$

$$R_{BB} = 6.546k\Omega$$

$$I_B = 6.67\mu A$$

$$I_{BB} = I_E$$

Handwritten calculations for Question 4:

$$I_C = 0.8mA$$

$$I_E = 8.08 \times 10^{-4}A$$



Course Name:	Electronic Devices & Circuits	Course Code:	EE225
Program:	Electrical Engineering	Semester:	Fall 201
Duration:	180 Minutes	Total Marks:	100
Paper Date:	18-12-2017	Weight:	50%
Section:	ALL	Page(s):	4
Exam Type:	Final		

Student : Name: _____ Roll No. _____ Section: _____

Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculation. Direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together. Take your assumption if any parameter is missing.

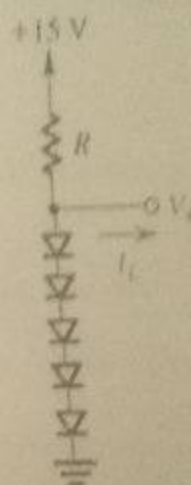
Question: 1 (CLO # 1)

10 Marks

Design the circuit shown in figure so that $V_o = 4V$ when $I_L = 0$ and V_o changes by 40 mV per 1 mA of load current. Find:

- The value of R
- The junction area of each diode relative to a diode with 0.7 V drops at 1mA current. Assume all diodes are identical and $n = 1$.

Hint: the junction area is directly proportional to saturation current I_s .

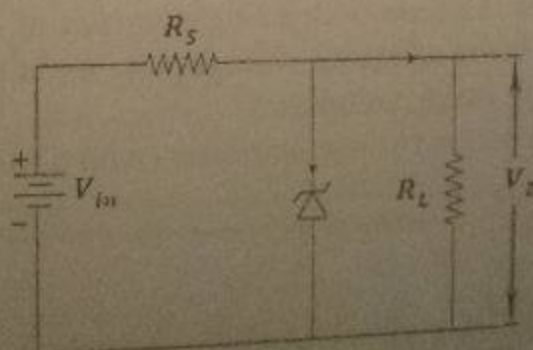


Question: 2 (CLO # 2)

15 Marks

a) Consider the following circuit

- Assume $V_{in} = 15V$, $R_s = 1k\Omega$, $V_z = 5V$ and $I_{knee} = 1mA$, Find the minimum value of R_L so that diode stays in breakdown region.
- Assume $V_{in} = 15V$, $R_L = 200\Omega$, $V_z = 5V$ and $I_{knee} = 1mA$, Find the maximum value of R_s so that diode stays in breakdown region.



Question # 5 [15 marks]

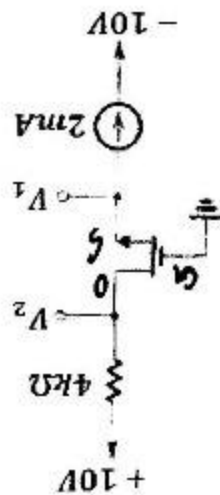
A common base amplifier is biased at $I_E = 0.25\text{mA}$ with $R_C = R_L = 10\text{k}\Omega$ and is driven by a signal source $R_{sig} = 1\text{k}\Omega$. Find the overall voltage gain G_v . If the maximum signal amplitude of the voltage between base and emitter is limited to 10mV , what are the corresponding amplitudes of v_{sig} and v_o ? Assume $\alpha \approx 1$.

Question # 6 [10+10=20 marks]

a) An NMOS transistor operating in the triode region with $v_{DS} = 0.1\text{V}$, is found to conduct $60\mu\text{A}$ for $v_{GS} = 2\text{V}$ and $160\mu\text{A}$ for $v_{GS} = 4\text{V}$.

- What is the apparent value of threshold voltage V_t ?
- If $k'_n = 50\mu\text{A}/\text{V}^2$, What is the device W/L ratio?
- What current would you expect to flow with $v_{GS} = 3\text{V}$ and $v_{DS} = 0.15\text{V}$?
- If the device is operated at $v_{GS} = 3\text{V}$, at what value of v_{DS} will the drain end of MOSFET channel just reach pinch off, and what is the corresponding drain current?

b) In the circuit shown in figure, transistor is characterized by $|V_t| = 2\text{V}$, $k'_n W/L = 1\text{mA}/\text{V}^2$. Find the labeled voltages V_1 and V_2 .



National University of Computer and Emerging Sciences, Lahore Campus



Course: Electronic Devices & Circuits
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 11-Nov-16
 Section: ALL
 Exam: Midterm-2

Course Code: EE225
 Semester: Fall 2016
 Total Marks: 40
 Weight: 15%
 Page(s): 2
 Roll No:

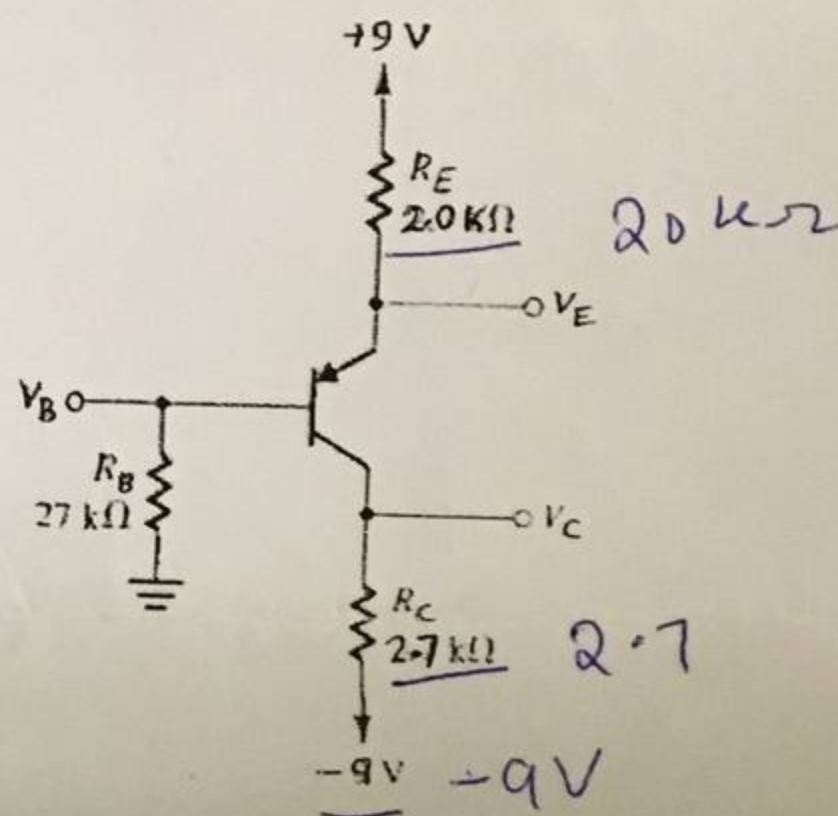
Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

In the circuit shown, the transistor has

$$\beta = 30.$$

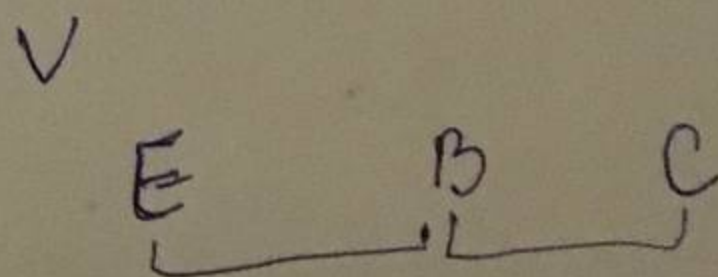
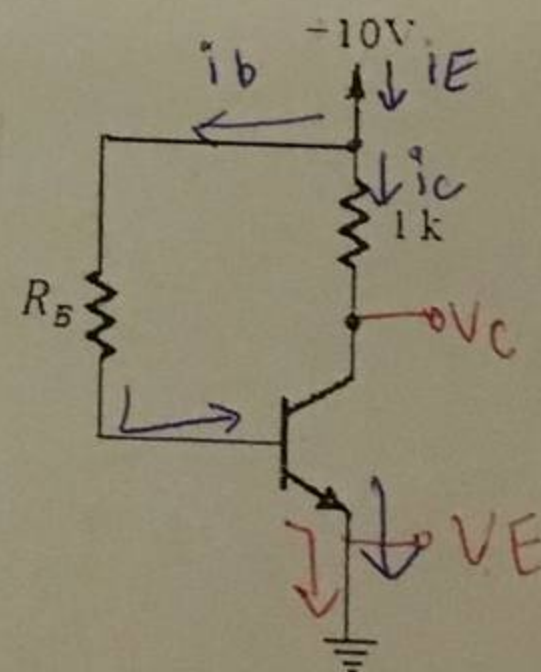
- Find the values of V_B , V_E and V_C ?
- Find the largest value to which R_C can be raised while the transistor remains in the active mode. - 21.07



Question # 2 [10 marks]

For the circuit shown in figure, select a value of R_B so that the transistor saturates with an overdrive factor of 10. The BJT is specified to have a minimum β of 30. What is the value of forced β achieved? 3

$$30 \times 3 = 90$$



Question # 3 [15 marks]

A common emitter amplifier utilizes a BJT with $\beta = 200$ and $V_A = 200$ V. It is biased at $I_C = 0.5$ mA and has a collector resistance $R_C = 10$ k Ω .

- Draw and label the small signal equivalent circuit using Hybrid- π model including source and load resistance.
- Find R_{in} , R_o and A_{vo} ?
- If the amplifier is fed with a signal source having a resistance of 10 k Ω and a load resistance R_L of 10 k Ω is connected to the output terminal, Find A_v and G_v ?
- If a peak-to-peak output ac voltage of 1 V is measured at the collector, what ac input voltage and current must be associated with the base?

EE225 - Electronic Devices and Circuits
Mid Term – I (FALL 2016)

Maximum Marks: 35

Date: 20-09-2016

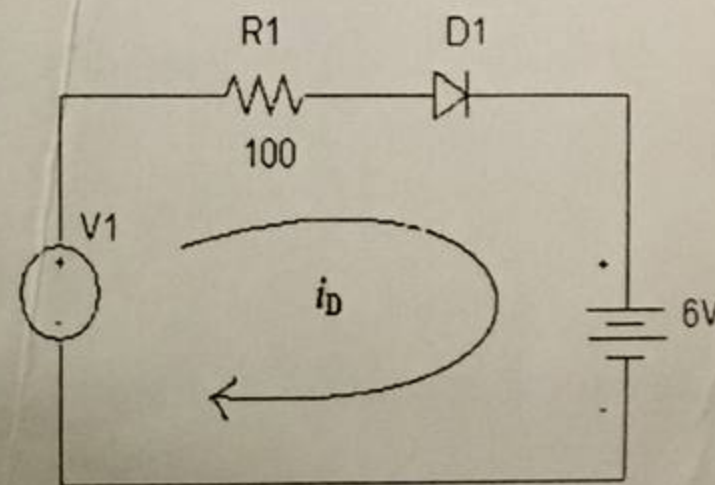
Total Time: 60mins

INSTRUCTIONS: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [3+2+3+4+3=15 marks]

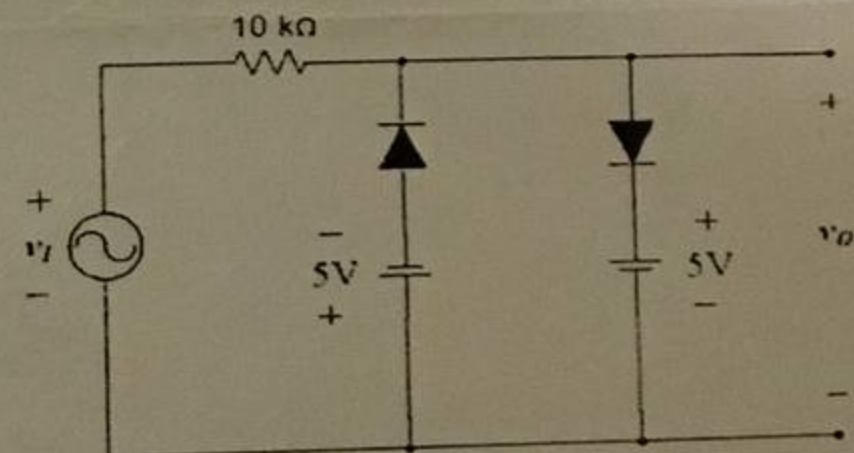
The charging circuit of a battery is shown in the figure, where the input V1 is a sinusoid with a peak voltage of 24 volts. Assume that diode is ideal.

- Sketch and label the voltage waveform across diode.
- Sketch and label the voltage waveform across resistor.
- Sketch and label the current waveform.
- Find the fraction of the cycle during which diode conducts.
- Find the maximum reverse bias voltage that appears across the diode.



Question # 2 [10 marks]

Assuming the diodes to be ideal, draw the transfer characteristics (v_o versus v_i) of the circuit shown. Take $v_i = 20 \sin \omega t$.



Question # 3 [10 marks]

Design the circuit shown in figure to provide an output voltage of 2.4V. Assume that the diodes available have 0.7V drop at 1mA of current. Take $n=1$.

